Farshore AI technical spike

One-page summary / September 30, 2026

Model and method

Llama 3.2 3B Q4_K_M, Ollama 0.21.0, local CPU. Intel Core Ultra 5 125H, 15.68 GiB RAM and 2.12 GiB free at start. Six sequential inputs through the existing adapter from a standalone script. Strict JSON schema, temperature 0.2, 450 output-token cap. No loaded model before S1. No repeated/concurrency trials.

Latency

Minimum 8.87 s, average 26.36 s, maximum 38.43 s. Individual wall-clock times: S1 38.43 s, S2 28.95 s, S3 8.87 s, S4 25.43 s, S5 29.38 s, S6 27.14 s. Includes first-call loading, request, parsing and validation. Failure timings excluded.

Output quality

All six replies passed structure validation and described actions matching safeChoice. S4/S5 take the requested action but report followed=false. S6 has no definite requested action but reports followed=true. Analytic review of six cases does not establish a production error rate. The archived September 23 reply/score contradiction remains relevant.

Failure behavior

E1: malformed JSON to model service, HTTP 400, unexpected EOF. E2: missing model, HTTP 404, model name not found. E3: deliberate 1ms client deadline, TimeoutError, no model response received. E4: one-character advice, app HTTP 400 before provider contact. Exact provider bodies and client errors are archived separately.

Limits and cost

No 429 in this sequential local run and no quota/concurrency threshold measured. Existing adapter handles 429. No billed inference API was used. Hardware/electricity, browser speech services and future cloud costs are unmeasured.

Effect on the design

Proposed elapsed wait and cancel for long requests. Retain counsel, retry and provide explicit Story mode. Review the action and rationale before score commitment, with refine/discard controls for uncertainty. No invented confidence percentages. Keep Llama provisionally for the prototype and evaluate decision flags.

Evidence and sources

evidence/spike-2026-09-30.json, evidence/quality-review.json and scripts/phase2-spike.mjs. Official Ollama compatible API and structured outputs documentation, accessed September 30, 2026. Proposed controls are design deliverables, not implemented game changes.